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A survey of image spamming and filtering techniques

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Abstract Many techniques have been proposed to combat the upsurge in image-based spam. All the proposed techniques have the same target, trying to avoid the image spam entering our inboxes. Image spammers avoid the filter by different tricks and each of them needs to be analyzed to determine what facility the filters need to have for overcoming the tricks and not allowing spammers to full our inbox. Different tricks give rise to different techniques. This work surveys image spam phenomena from all sides, containing definitions, image spam tricks, anti image spam techniques, data set, etc. We describe each image spamming trick separately, and by perusing the methods used by researchers to combat them, a classification is drawn in three groups: header-based, content-based, and text-based. Finally, we discus the data sets which researchers use in experimental evaluation of their articles to show the accuracy of their ideas.

Keywords Image spam · Image classification · Spam filtering techniques

1 Introduction

At the current state of the world, thousand of million people are connected together through electronic devices and they are involved in high volume and various collaborations around the world. Nowadays the common and popular application which is used for transferring messages is Electronic mail. Email service is very cheap, take no time, user can send and receive just in real time, remove distance problems, best for official communications, removes time-zone limitations, and etc. Beside the all mentioned advantages, the problem of undesired

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electronic messages is nowadays a serious issue of email services and encountered various problems over the time.

Some people and companies exploit such popular and free service to achieve their aims, such as advertising. During transferring legitimate emails, batch of unsolicited emails which are called spams are transferring for advertisement aim. Exceeding of sending spam by spammers has become an important problem. Spam cause a lot of inconvenience for both servers and users, we list troubles separately for server and user in the following:

- Server: High volume of data follows in communication directions cause delay and/or missing in service responsency (Uemura and Tabat 2008), decline credibility and reliability of email service (Uemura and Tabat 2008), and use up supplied storage of server (Soranamageswari and Meena 2010; Stuart et al. 2004).
- User: Everyday users receive so many emails which takes time to classify and separate authentic ones, necessity of eliminate phony emails, threat user security strongly (for example, by clicking a link), compromised computers and identity theft (Network security articles for windows server, available at <http://www.windowsecurity.com>).

According to mentioned problems, presence of an efficient method which is able to distinguish spam from legitimate email is necessary.

In the second half of the last decade, email service encounter image spams as a new threat which is the most sophisticated kind of spam emails up to now. It is the most important one because it make message interesting for user and hard to detect. In earlier year spammer put the messages in body of the emails; it is a traditional way for spammers. There are a wide number of text based anti-spam filters that overcome the traditional way of spamming (Soranamageswari and Meena 2010; Wang et al. 2007). Today spammers circumvent images in three ways, embedding in email body, attaching an image file to emails and placing a hyprelink in the body of the email and the target message is carried by the image. In fact, spammers use this kind of spams to disable traditional text-based filtering techniques.

Image spams are rich in content and variety of kind, so image spam filters apply different methods on their filters to be able to detect image spam aiming to battle the image spammers. The simplest method is to use OCR (Optical Character Recognition) techniques to extract text of image spam and analyzing the text. Some of them are using feature extraction methods to analyze images.

1.1 Image spam statistics

Image spam rapidly become known as a mean to easily evade any textual analyzer embedding in the different spam filters (Zhen et al. 2009; Kelly 2007). The technique of including an image instead of text in spam emails started in 2004 (Kelly 2007) so that up to late 2005, 1% of all spam emails were image spams (Soranamageswari and Meena 2010). In 2006 and 2007 a great growth occurred so that 27% and 65% of all spam emails are reported as image spam, respectively (Mehta et al. 2008; Gao et al. 2008). This upsurge means that image spam firmly established its generality among cybercrime. Since the researchers have tried a lot for detecting image spam emails among the legitimate emails, the volume of image spam decreased in 2008 and 2009, roughly to 40% of all spam emails (Zuo et al. 2009). Although, because of the new spam tricks, image spammers won the battle again and image spams are again on the rise, so that the image spam emails as reported in Symantec (Antivirus, Anti-Spyware, available at <http://www.symantec.com>) are 55% of spam emails in 2010. These statistics encourage the researchers to pay attention to image spam more than before and it will be more considerable by knowing that the spam emails are 85% of all emails

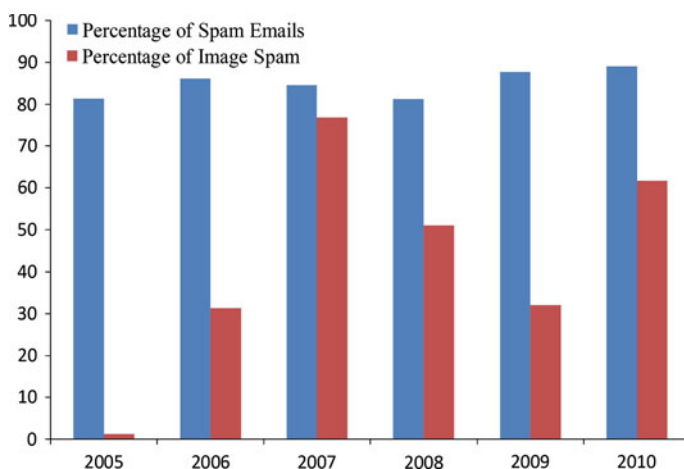


Fig. 1 The annual report of SYmantec (Antivirus, Anti-Spyware, available at <http://www.symantec.com>) on the percentage of image spams and spam emails in emails

in 2010 (Secure Web Gateway—Internet Security and Email Security Solutions, available at <http://www.m86security.com>). Figure 1 shows the percentage of image spams in all of the exchanged emails which is reported by SYmantec (Antivirus, Anti-Spyware, available at <http://www.symantec.com>).

1.2 Motivations

The main motivations of this work can be summarized as follows:

- Spam and its new emerged namely image spam target most popular way of communication i.e. Email in recent years.
- Image Spam email is more destructive than ordinary spam emails and needs more attention.
- Spam specially image spam is becoming popular and more popular for researchers in recent years, there are a lot of surveys about spam but none of them addresses image spams.

1.3 Contribution

The main contributions of this work can be summarized as follows:

- The absence of image spam survey issue is addressed by analyzing all of works done in image spam field so far.
- This work is not only a survey but also author's ideas, innovations, and discussions on the Image spam issues.
- Vulnerability of all three groups of anti image spam techniques is presented.
- In image spam research each author uses its own data set. There is not a standard data set to evaluate filtering methods. In this work all data set exploited by researchers are analyzed in details and a mechanism to constitute a universal data set proposed.
- This work has scrutinized completely a classification of all tricks which exploited by image spammers and four effective and novel tricks are proposed which were not mentioned in the literature before.

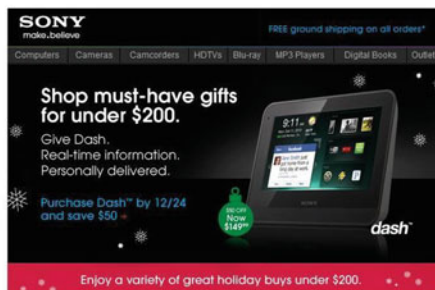
The rest of the paper is structured as follows: a general description of image spams is presented in Sect. 2. Section 3 describes the image spam tricks. Anti image spams are presented in Sect. 4. Section 5 describes the data set used by researchers and evaluation criteria and finally open issues of image spam tricks and their detection techniques are discussed in Sect. 6.

2 Image spam

This section provides an introduction to the phenomena of image spams. Spammers took advantage of images in order to battle with text-based filtering techniques. Image spam can easily escape traditional spam filters and becomes more and more difficult to detect (Qu and Zhang 2009). Spammers use image in a spam email for embedding their target messages (Figs. 2, 3).



(i)



(ii)



(iii)



(iv)

Fig. 2 Samples for (i) financial, (ii) products, (iii) internet and (iv) leisure image spams



Fig. 3 Samples for (i) health and (ii) education image spams

2.1 Definition

There exist various definitions of what image spam emails are and how they differ from other emails. Image spam has different meaning for different people and there is no uniform definition in theory up to now. Image spam is defined as: the actual spam message is moved into an image attached to the message (Wang et al. 2007; Nhung and Phuong 2007; Krasser et al. 2007; Gargiulo and Sansone 2008), image spam means spam including image which appears in the main body or attachment the main body (He et al. 2009), image spam is a kind of email spam where the message text of the spam is presented as a picture in an image file (Soranamageswari and Meena 2010; Zhen et al. 2009), the image is a hyperlink to an unknown web page (Klangraphant and Bhattarakosol 2010; Mehta et al. 2008; Aradhye et al. 2005) and some other ideas which are quite similar to the mention definitions.

2.2 Image spammer motivations

Basically the main motivation behind the image spam is circumvent the spam filters. Image spam is an attempt by spammers to hide their message from anti spammers (Nielson et al. 2008; Zuo et al. 2009). Spammers send their messages in attached images that are readable by human but hidden from a text-based filter (Gargiulo and Sansone 2008). By emerging image in spam email, spam email filters encountered a new trouble and they had to spend a lot to solve this important communication problem. Exploiting image in spam email is too expensive for spam filters. Clearly, image processing is more expensive than text processing. Identification of image spam requires content analysis of the attached image. Since images are a large and complex data format, in order to successfully identify them, it is needed to have a deep comprehension about the properties of image spams (Mehta et al. 2008). Image spammer usually take the advantages of images to achieve their spamming goals. As you heard the well known sentences, "A picture is worth a thousand words", a complex idea or concept can be conveyed with just a single simple image almost all times. So, image spammers use images for making their spam emails more attractive and more expressive.

2.3 Types

Image spam emails are seen in different spam emails with different shapes. To our knowledge, image spam emails are usually classified into three branches:

- Image spam content contains an image which presents the spammer target and an URL that address the image spammer website. In this circumstances user after receiving the mail should type the URL in address bar to visit the website. So the exploited image should be attractive enough to persuade the user to do that.
- Image spam content is similar to all content which are used in text based spam email. It can be said that the used image is a screen shot of the usual text based spam email. All targets are seen in the image with all details that spammer want to share for users. For example if spammer want to show ads in image spam, it can contain product name, description of the product, producer name, address, telephone number, and etc.
- Image spam is a hyperlink to a website. User after clicking on image can see the special website that contains all description of spammer target with complete details. Because of user curious, the hyper linked website usually opened by a simple click on the image.

Usually image spam email target different advertisement aims which are listed below:

Adult: Email attacks containing or referring to products or services intended for persons of legal age (above the age of 18), often offensive or inappropriate. E.g. pornography, dating services, personal ads and relationship advice.

Financial: Email attacks that contain references or offers related to money, the stock market or other financial opportunities. E.g. investments, credit reports, real estate and loans. Fraud image spams are also parts of Financial image spams. They are email attacks that appear to be from a well-known company, but are not. Also known as brand spoofing or phishing, these messages are often used to trick users into revealing personal information such as email address, financial information and passwords. E.g. account notification, credit card verification and billing updates. Scams are also as part of financial image spams. They are email attacks recognized as fraudulent, intentionally misguiding, or known to result in fraudulent activity on the part of the sender. E.g. Pyramid schemes and chain letters.

Products: Email attacks offering or advertising general goods and services. E.g. manufactured goods, services, investigation services, clothing and makeup.

Internet: Email attacks specifically offering or advertising Internet or computer-related goods and services. E.g. web hosting, domain name registration, email marketing, web design.

Leisure: Email attacks offering or advertising prizes, awards, or discounted leisure activities. E.g. vacation deals and online casinos.

Health: Email attacks offering or advertising health-related products and services. E.g. dietary supplements and sport, disease prevention, pharmaceuticals, medical treatments and herbal remedies.

Political: Email attacks that advertise a political candidate's campaign, request donations to a political party or cause, offer products related to a political figure/campaign, etc. E.g. political party, elections, donations.

Education: Email attacks offering a fortune related to education. E.g. e-learning, institute, get a degree, Continuing education.

Spiritual: Email attacks that contain information pertaining to religious or spiritual evangelization and/or services. E.g. psychics, astrology, organized religion, outreach Other image spam emails attacks not pertaining to any other category.

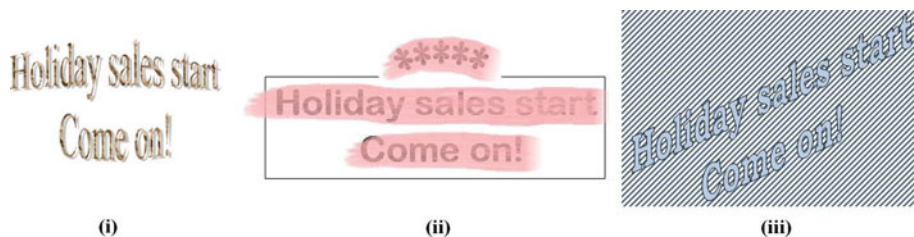


Fig. 4 Samples of obfuscation used in image spam

3 Image spam tricks

There are a lot of tricks which spammers exploit to make anti-spam methods unprofitable. Each time that spammers employ a trick, anti-spam researchers try to overcome the trick and restore efficiency. This battle continues during the time, again spammer try new wreckful trick and this story repeats. In the rest of paper we plan to introduce and analyze all important tricks that used from earlier years up to now. All tricks are categorised as three classes: ordinary tricks, bold tricks and new proposed tricks.

3.1 Ordinary tricks

3.1.1 Obfuscation

Obfuscation is one of the oldest tricks in this field, e.g. misspelling words, rotating the text slightly, blurring of text outlines, add shadow, and adding random noise to make image hard to read and prevent the detection of multiple similar images. Today this trick is well-known and anti spam filter can almost completely recognize text. Figure 4 shows three samples of obfuscation.

3.1.2 Text only

Some image spam comprise only text (Fig. 5), for detecting this kind of spam as usual text analysis is needed, but first we should extract text from image. But it is clear that this situation needs very high memory and processing elements which makes it so expensive.

3.1.3 Multiple sections

This trick splits original image into multiple sections (Fig. 6). Since some anti-spam approaches employ signature filtering, they encounter significant problems to overcome this trick and it makes text reorganization hard. As an example parts of one word appear into multiple images. Email server reassembles sections when it is opened. Multiple sections technique is effective against many anti-spam solutions which use signature concept, because spammers send out multiple versions of the same image by slicing it randomly and then reassembling it within the email.

3.1.4 Hand-written images

Use of hand-written font (Fig. 7) is another trick. In this way they try to keep text hidden from anti spam systems that employ OCR tool. The image spam can simply bypass filtering

Hello,

You have been chosen to participate in an invitation only limited time event!

Are you currently paying over 3% for your mortgaged STOP! We can help you lower that today!

Answer only a few questions and we can give you an approval in under 30 seconds it is that simple!

And stop fighting for lenders, let them fight for you! Make them work for your business by giving you the lowest rates around! Two hundred and thirty thousand dollar loans are available for only three hundred and forty dollars/month!

WE ARE PRACTICALLY GIVING AWAY MONEY!

Think your credit is too bad to get a deal like this? THINK AGAIN! We will have you saving your money in no time!

Are you ready to save your money?

Regards

Fig. 5 Sample of text image spam

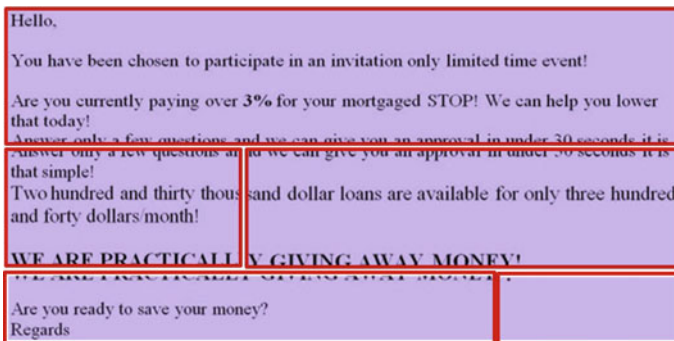


Fig. 6 Sample of multi section image spam

because hand-written font is not similar to any standard font that is recognizable by OCR tools.

3.1.5 Wild background

Spammers use images with vague and strange backgrounds. They use geometrical figures and various colors (Fig. 8). This trick target OCR because OCR is normally based on the measuring the geometry and it search for geometrical shapes which are similar to letters and put them into a text file. Any methods that employ OCR is a target of this trick.

3.1.6 Geometric variance

Many filters act based on sameness. Image spam can be altered easily without disturbing the message inside them. Thus one spam message will appear as dozens of the different shaped

Fig. 7 Sample of hand-written image spam

URGENT INVESTOR ALERT!!!
MONDAY AUGUST 28, 2006
GAMING TRANSACTIONS INC.
- ONLINE CASINO GAMES
- INTERNET GAMING PORTALS
STOCK SYMBOL: GOTS
TRADING AT: LESS THAN \$0.10 !!
5-DAY TARGET: \$0.75
CURRENT RATING: STRONG BUY!!
THIS STOCK WILL EXPLODE ON MONDAY!

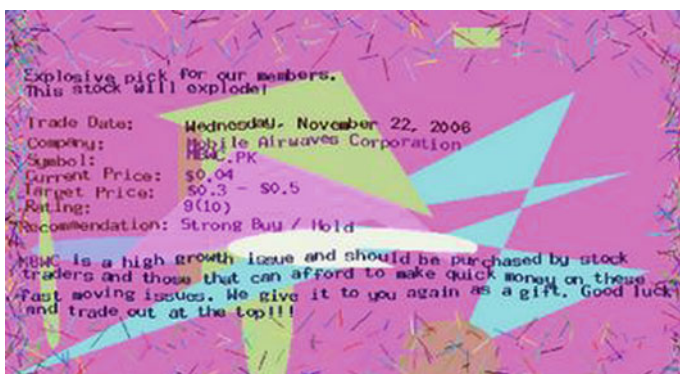


Fig. 8 Sample of wild background image spam

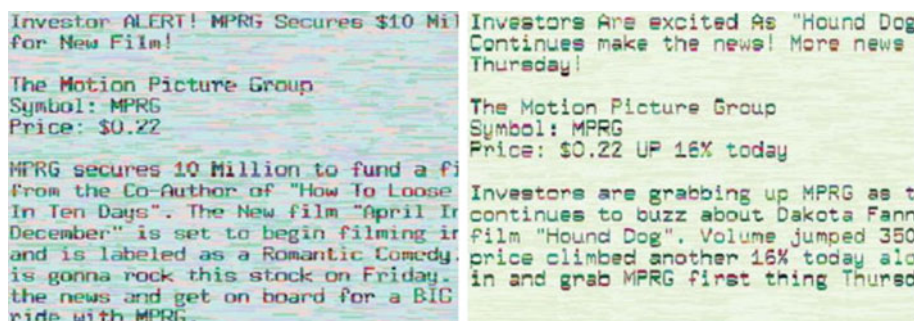


Fig. 9 Sample of geometric variance image spam

images, and each time the colors of the text images or background will be changed. In Fig. 9 there are two identical message in content shown, but they are different in properties.

3.1.7 Multi-frame animated Gifs

Another trick employed is the use of animated gifs. Some examples of the multi-frame animated gifs are shown in Fig. 10. These are created by combining multiple gif images in one

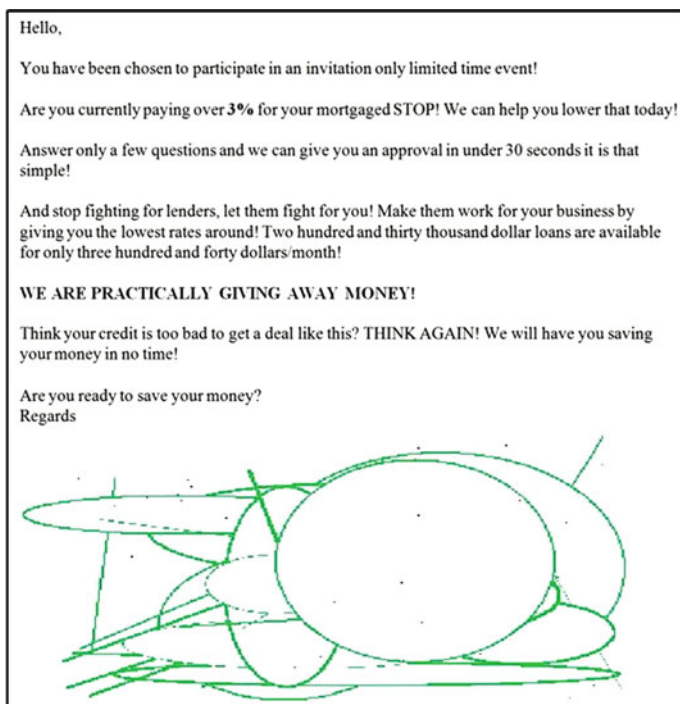


Fig. 10 Sample of multi-frame animated Gifs image spam

file, displayed one after another, gives the appearance of movement. In this regard instead of sending a singular image spam frame, spammer add some other frames to image spam frame. The extra frame(s) can be even blank.

3.2 Focus on bold tricks

Now we want to focus on some bold tricks which others have paid less attention to them. In the rest of the paper some neat tricks are presented which almost all the presented approaches so far will face trouble detecting them. These kinds of tricks can bypass both of image content-based and OCR-based techniques because of their special characteristics. This paper proposes three new and effective tricks that can be used by spammers in the future, “You can unsubscribe!”, natural image and patchy font.

3.2.1 Cartoon

This new trick use cartoon fonts which are unusual and odd to anti spam filter (Fig. 11). They use multiple colors, multiple styles, and special shapes so that produce a very beautiful artifact that is appealing and perceptible for human, but very hard to detect for machine.

Fig. 11 Sample of cartoon image spam



3.2.2 Recoloring

Spammers use same pictures with same objects but different colors (Fig. 12), this trick can strongly harass anti-spam methods. Recoloring can disable a lot of criteria that anti-spam methods exploit for filtering. Such as color histogram, pixel analysis, image signature, pixel average and etc.

3.2.3 Obfuscate and shorten URLs

Some URLs are known for filters regarding to their history of activity. Shorten URLs service allow spammer to bypass such filters by a different URL redirected users to spam sites. There are a lot of sites which provide this service for user in free. Some famous ones, including (<http://tinyurl.com/>; <http://ow.ly/>; <http://goo.gl/>; <http://bit.ly/>). Spam URLs sometimes are strange and unusual, some shorten URLs services provide optional feasibility. User can suggest what string the long URL should be convert to, so spammer can use some positive words for attract people trust instead of the words that reflect real concept of spam. As a first glance it seems checking final URLs by shorten URLs services can be efficient. But as pointed in (<http://research.zscaler.com/2011/02/unchecked-redirection-url-shortener.html>) spammer can exploit another tactic to escape from being blocked. The idea is based on the use of trusted intermediate site insted of the direct addressing. As an example <http://www.fmcsa.dot.gov/> is used by a pharmacy canadapharm.org at aiming redirection as reported in (<http://research.zscaler.com/2011/02/unchecked-redirection-url-shortener.html>).

3.2.4 Forge header information

Spammer can easily forge information in the header part of an email, so information obtained by header part cannot be reliable. Spammers do anything to keep their identity secret and only IP address of sender cannot be manipulated by spammer. For example spammers forge From:, Reply-To: and Return-Path: lines because they just think about money and do not like to receive complaints. And also they forge X-Mailer: for example they mix numbers randomly for the version of Outlook, it is obvious that this is not a real version at all.

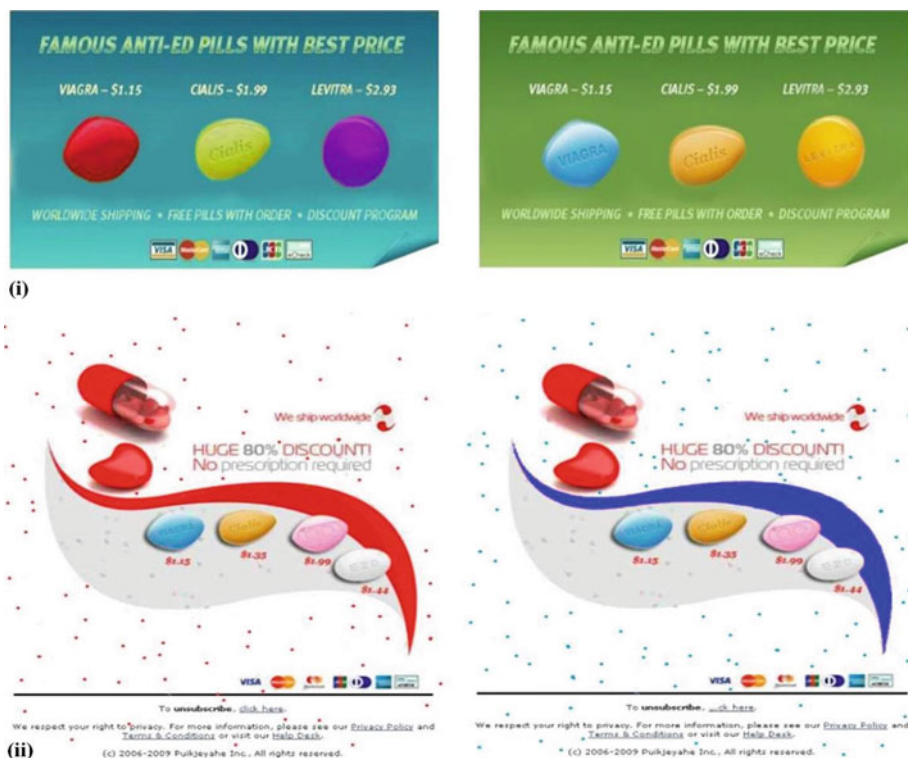


Fig. 12 Sample of recoloring image spam

3.2.5 Template-Driven

Spam generation tools produce spam in a predefined structure as pattern (Stern 2008). This structure is a constant framework; the things changing are only elements and contents of an email.

- Header: Spammers separate header from body, maybe there is no relevance between body and header. It causes to pass signature based anti spam successfully (Stern 2008).
- Body: Spammer use same template for body of email and various content. The concept and meaning play a critical role while words and sentences are unimportant. For example spammer use a certain template for body, but the words and sentences can be changed with a synonym.

3.2.6 Emerge new source Stone (2006)

Nowadays, there are a lot of spam sources that are known for companies and institutes. Each time that a spam source enterprise to send junk email, anti spam systems check out history and reputation of sender so can recognize the sender simply and block it. During the time new spam sources come out and anti spam system encounter some new problems. In this circumstance, it is clear that anti spam systems cannot rely on black list.

3.2.7 People's computers

As reported in Stone (2006) the spammers try to infect people's computers so that use people's accounts and contacts list. By sending desirable emails through personal accounts, spammers address two major goals. First their identity keeps unknown for users and companies. Second they discover new destination accounts for their attacks: contact list of people.

As mentioned in Stone (2006) Daniel Drucker, a vice president at the antispam company Postini said "Because they are stealing other people's computers to send out the bad stuff, their marginal costs are zero and the scary part is that the economics are now tilted in their favore." Here is one of the practical examples which reported by eSecurity Planet (www.esecurityplanet.com/) in Dec. 2010; victims think the email with the "Here You Have" subject line have relevance to pornographic videos or photos while when they open these unsolicited emails an SCR executable file that disabled their security software apps and start to work. It's mission is sending the same spam message to all the contacts in their address books.

3.2.8 Harvest email addresses

There is a lot of spam bots over the World Wide Web to collect email addresses and find new destinations to attack. It searches for any string with format of X@X.X like test@example.com. Today many people obfuscate their email address on the web like this: "test [at] example [.] com". While it seems efficient, but it does not work properly because email address format and symbols and letter can used in are limited, machine work well on limited environments so some email obfuscation decoder was designed like one in [<http://jasonpriem.com/obfuscation-decoder/>].

3.3 Four new and effective tricks

3.3.1 You can unsubscribe!

Spammers do anything to have more contact with you, they strongly persuade you for something more than receiving and reading an email, for example make a phone call or visit a webpage or something else. You may encounter such sentences in a spam image "If you would like to change your email address in our database or opt-out, please click here" or "To unsubscribe from e-mailing click here". There is one big question here, do you ever subscribe to this site?!, of course no! Figure 13 shows an example.

3.3.2 Natural image

Spammers use natural images (Figs. 14, 15) for embedding their messages in it. Feature of natural images cause to disable those kinds of image spam filters which based on special characteristics of spam images. Because postiche images are usually so different in comparison with naturals.

3.3.3 Patchy font

In this trick each character is composed of many stuffs. OCR mainly use two methods for recognition, matrix matching and feature extraction. Matrix matching is used in older OCR tools or in a bounded environment which includes limitation of character and styles. Feature extraction does not rely on precise matching, it analyze features such as open areas, closed

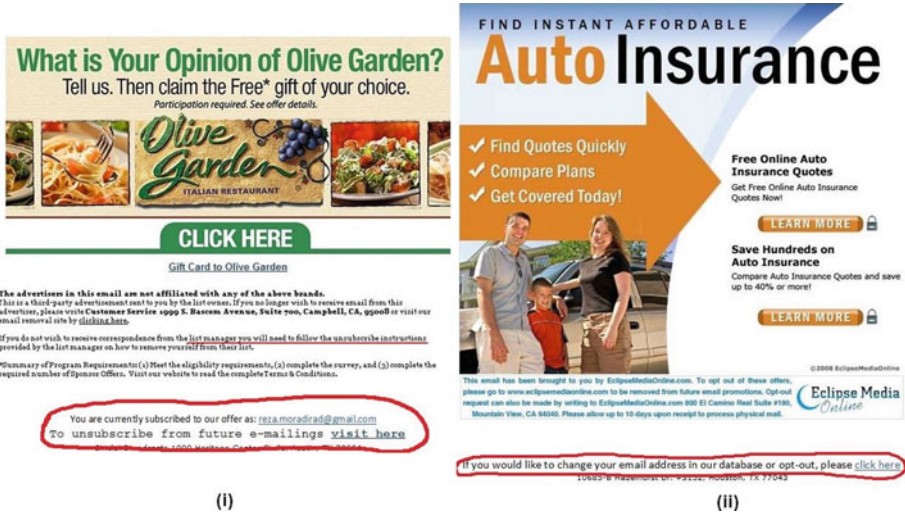


Fig. 13 Sample of *you can unsubscribe!* image spam

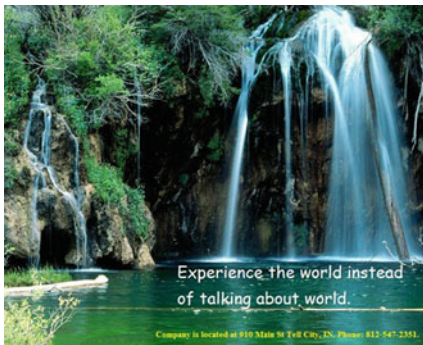


Fig. 14 Sample of natural image spam

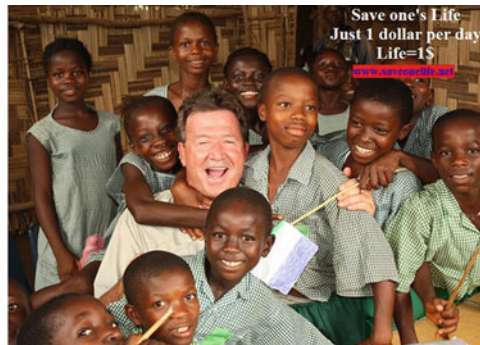


Fig. 15 Sample of patchy font based image spam



shapes, diagonal lines, line intersections, etc. (Jithesh et al. 2003). It is obvious that against of this trick matrix matching methods are fully ineffective, just consider if any feature extraction methods in future will be able to recognize (note that it is so sophisticated therefore so hard to recognize), we suggest the right one of following figure, in this condition it find phony character. For example in character “S” in following figure, it will find “BIGSAL..”.

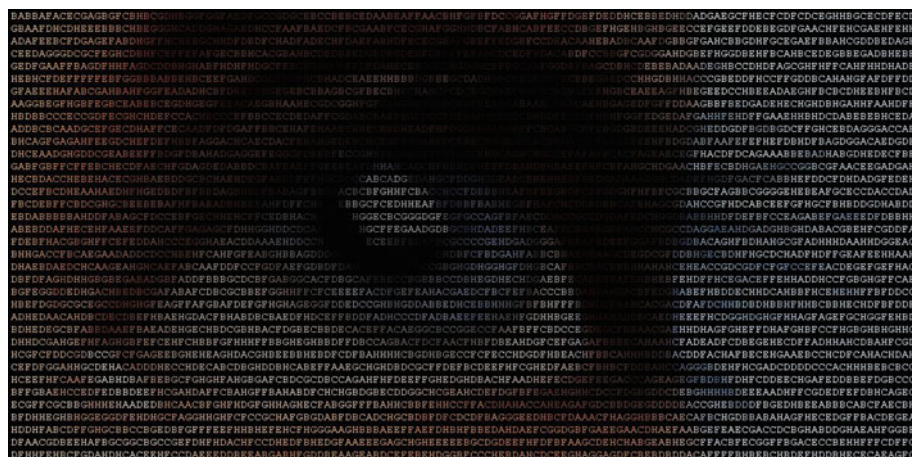


Fig. 16 A sample of a text image spam

3.3.4 Text image

In this trick spammer exploit alphabets to draw their special intent, please note that in text-image emails no image file like jpg, png, gif, tiff and etc are appeared. Spammer uses no images but use inherent concept of images to transmit messages. A sample of a text image spam is shown in Fig. 16.

4 Anti image spam

After the evolution and distribution of text based filtering techniques, the spammers adopted the use of image spam. The text or the target of an advertisement is embedded in an image, so that it is impossible to analyze the email content with traditional plain text based filters. This led to the need for image analysis based filters. In image based filtering the main issue is to find more performance algorithms to detect image spam email from non image spam email. In addressing this growing problem, many techniques as solutions to image spam reduction have been proposed. Image spam filtering techniques are exploited to separate spam images from other common categories of e-mail images. All image spam filtering techniques belong to three main groups:

- Header based techniques eliciting the spam email properties for analyzing and detection.
- Content based techniques utilizing feature extraction and image content analysis.
- OCR based techniques utilizing OCR (Optical Character Recognition) and process text.

In the rest, all three groups are described individually.

4.1 Header based

Today modern email clients often hide the headers from user view that is why many people have never seen an email header. However, headers are always delivered along with the message contents. Most email clients provide an option to enable or disable display of email header. The idea of this group is analyzing only header part of emails because E-mail messages contain more than just a message. Header part of email consists of many fields

which provide useful peripheral information. Just some of header fields listed here: Subject, Date, From, To, Cc, Bcc, References, Comments, Keywords, Sender IP, Sender email address, Precedence, List-Help, Errors-To, Sender, In-Reply-To, Delivered-To, X-Mailer, X-Priority, Delivered-To, X-MimeOLE, Content-Transfer-Encoding, Message-ID, List-ID, MIME-Version, Importance, Incomplete-Copy, Priority, Sensitivity, Language, Conversion, Message-Type, Fax, Telefax, User-Agent, Originator-Info, Phone, X-Loop, Status. Some of the mentioned header fields are used in practice and they are described below:

- *Sender IP*: Every computer connected to the Internet is assigned a unique numerical code known as an IP address (Internet Protocol Address). IP addresses allocation is two types static and dynamic. Sender IP address indicate address of the source computer from which the email was sent. It can tell where the email was sent from and possibly who sent it.
- *Sender email address*: Sender email address is a certain string specifies the sender email account.
- *Precedence*: Email precedence is a label in email header part that indicates email type; emails can be delivered to the list, bulk mail, junk mail or in a custom way. Many anti spam systems in both email servers and email clients of end user pay attention to precedence label to manage email messages fitness.
- *List-Help*: List-Help indicates an email URL and/or web URL to obtain help for a list. It may also include some “comments”. The List-Help also considered as the most important field in email header part.
- *Errors-To*: Errors-To indicate the address which error notifications are to be sent. In other hand it’s request to get delivery notifications. If errors occur anywhere during processing, this header field will cause error messages to go to the listed addresses. Errors-To field will go away in a future release.
- *Sender*: Sender specifies the mailbox of the agent responsible for transmission of the email message.
- *In-Reply-To*: In-Reply-To specify message identifier of the original message to which the current message is a reply to.
- *X-Mailer*: X-Mailer provides information about the client software of the originator in other hand describe the software used to creating and sending the message. For example, if you send the email using Outlook, the X-Mailer header field says Outlook and it’s version.
- *X-Priority*: X-Priority specifies the messages priority. The field is given a numerical value of 1 through 5: Values: 1 (Highest), 2 (High), 3 (Normal), 4 (Low), 5 (Lowest). 3 (Normal) is default if the field is omitted.
- *Delivered-To*: Delivered-To specifies recipient address.
- *X-MimeOLE*: X-MimeOLE added by Microsoft Outlook and possibly other Microsoft software.
- *Content-Transfer-Encoding*: Coding method used in a MIME message body part.
- *Message-ID*: Message-ID field is a unique identifier of each particular version of email message. It is like the tracing ID of an express postal mail. Message-ID is composed by the name of the server that assigned the ID and a unique string.
- *Date*: Specifies the date and time at which the creator of the message indicated that the message was complete and ready to enter the mail delivery system. The time when the message was written (or submitted)
- *From*: Specifies the author of the message, the mailbox of the person or system responsible for the writing of the message.
- *To*: Contains the address of the primary recipient of the message.

- *Received*: Received contains information about receipt of the current message by a mail transfer agent on the transfer path.

The paper [Saraubon and Limthanmaphon \(2009\)](#) presents a spam filter by analyzing the email header, its authors claim that it works well with both text-based spam and all kinds of image spam. They use only Sender IP address and Sender Email address. They are able to specify the country that the IP address belongs to and also what country that MX hosts of the sender are located in. The paper [Liu et al. \(2010b\)](#) present a method which includes three layers of processing. The first layer only analyzes the mail header part, the second and third layers analyze the high level features and low-level features of images. They believe that some header fields appear in normal (or spam) mails with high probabilities, but appear in spam (or normal) mails with low probability and inverse. They build a set of 34 header fields so that percentages of these fields appearance between ham and spam are more than a certain threshold. These header fields include: Precedence, List-Help, Sender, Errors-To, X-Mailer, In-Reply-To, X-Priority, Delivered-To, X-MimeOLE, Content-Transfer-Encoding and some others. The authors claim that they achieve 93.7% in determination of image emails successfully. The paper [Krasser et al. \(2007\)](#) use just the width and the height denoted in the header of the image file, the image file type, and the file size. They use C4.5 decision tree and support vector machine classifiers with aimed at achieving high performance while their method is inexpensive because these features easily can be obtained from header part. The paper [Stuart et al. \(2004\)](#) analyzes some features from message subject header that we list here:

1. Number of alphabetic words that did not contain any vowels
2. Number of alphabetic words that contained at least two of the following letters (upper or lower case): J, K, Q, X, Z
3. Number of alphabetic words that were at least 15 characters long
4. Number of tokens that contained non-English characters, special characters such as punctuation, or numeric digits at the beginning or middle of the token.
5. Number of words with all alphabetic characters in upper case
6. Binary feature indicating occurrence of a character (including spaces) that is repeated at least three times in succession: yes = 1, no = 0.

And also they pay attention to X-Priority and content-type and finally apply neural network classifier. The paper [Ye et al. \(2008\)](#) also is header based and present a fully header based method which analyze Return-Path, Received, Message-ID, From, To, Date, X-Mailer. They use Support Vector Machine for classification.

4.2 Content based

These kinds of filters analyze image content and extracting some features which represents a property of the entire spam image. These filters use features which can be elicited from image efficiently, to represent major properties of the image. In content based classification tasks, effective feature is absolutely necessary because classifiers will not achieve accurate results without the perfectly represented features ([Cheng et al. 2010](#)). So far, researchers have proposed a number of image features, which can be used for characterizing certain aspects of images. Also image spam has been extensively studied using several techniques primarily developed from the Image Processing and Computer Vision community, using features of image ([Gargiulo et al. 2009](#)).

Since spam images have special characteristics ([Nhung and Phuong 2007](#); [Cheng et al. 2010](#)) which make them different from non-spam images, it is desirable to use features able to

capture such characteristics. Spammers use different randomization and obfuscation methods (Wang et al. 2007) to introduce spam image as non-spam image, and because of that none of the features cannot be used only as a feature for classification. So spam image filter techniques usually exploit several image features for classification. Features which usually exploited by image spam filter techniques are listed below with their definition and the techniques which use them (as a reference in bracket).

4.2.1 Color

The goal of using color feature is to obtain compact, perceptually relevant representation of the content of an image. Color features are among the most important and extensively used low-level features in image spam detection. They are usually robust in noise, resolution, orientation, resizing, and etc. Due to their little semantic meaning and its compact representation, color features tend to be more domain independent compared to other features. Color features combined together and can be used to describe the appearance of a special image. All color features which used by researchers who study in image spam field are listed here.

- *Color Saturation*: Color saturation is used to describe the intensity of color in the image. It is quantified as total number of pixels in the image for which the difference $\max(R, G, B) - \min(R, G, B)$ is greater than some threshold T (Krasser et al. 2007; Aradhye et al. 2005; Dredze et al. 2007; Liu et al. 2010a,b; Wang et al. 2010).
- *Color Histogram*: Color histogram is a representation of the distribution of colors in an image. A color histogram represents the number of pixels that have colors in each of a fixed list of color ranges. For a given image, the color histogram is a compact summary of the image. A color histogram is a vector $(h_1, h_2, \dots, h_n)$, in which each bucket h_i contains the number of pixels of color i in the image. (Soranamageswari and Meena 2010; Mehta et al. 2008; He et al. 2009; Gao et al. 2008; Chen and Zhang 2009; Wang et al. 2007; Gao et al. 2009, 2010; Gao 2009; Zhen et al. 2009).
- *Gray Histogram*: It is a statistical presentation of gray image. Global histogram contains the frequency of all gray color values in image. Gray histogram defined as $H(k) = \frac{n_k}{N}$, so that k is the gray-level of gray image, n_k is the number of pixels which the gray-level is k and N is the total number of image pixels (He et al. 2009; Liu et al. 2010a,b; Wang et al. 2010).
- *Entropy of the Histogram*: After obtaining image histogram, Entropy of Histogram can itself be a feature (Gao 2009; Gao et al. 2009, 2010).
- *Color Moment*: Color moments are measures that can be used to differentiate images, based on their features of color. Once calculated, these moments provide a measurement for color similarity between images. These values of similarity can then be compared to the values of images indexed in a database. The use of color moments is based on the assumption that the distribution of color in an image can be interpreted as a probability distribution. Probability distributions are characterized by a number of unique moments. It uses three central moments of an image's color distribution; they are mean (average color value in the image), standard deviation (square root of the variance of the distribution) and skewness (measure of the degree of asymmetry in the distribution) (Mehta et al. 2008; Qu and Zhang 2009; Wang et al. 2010; Zhen et al. 2009).
- *Contrast Ratio*: The contrast ratio is a property, defined as the ratio of the luminance of the brightest color (white) to that of the darkest color (black) (Gargiulo and Sansone 2008; Gargiulo et al. 2008).

- *Color Coherence*: Color coherence of an image is the degree to which pixels of that color are members of large similarly-colored regions. We refer to these significant regions as coherent regions, and observe that they are of significant importance in characterizing images. The measurement is based on classifying the pixels to test whether or not they are coherent. Coherent pixels are a part of some sizable contiguous region, while incoherent pixels are not (Mehta et al. 2008).
- *Random Pixel Test*: It produces n random colors and check to find if they appear in the image. A non spam image is more likely to have a wider range of colors (Dredze et al. 2007).
- *Color Heterogeneity*: The original color image is scaled by the maximum possible intensity such that the intensities in the *RGB* channels are within the range $[0, 1]$. It is then converted to an indexed image using minimum variance quantization such that the number of colors in the indexed image. The *RMS* (root mean square error) errors between the original image and the indexed image are then calculated individually for the text and non-text parts of the image, which form our two color heterogeneity features (Aradhye et al. 2005; Zhen et al. 2009).
- *Energy*: The spectral content of an image (Gargiulo and Sansone 2008; Gargiulo et al. 2008).
- *Prevalent Color Coverage*: It can be evaluated by counting the number of pixels with the most common color appearance in the image. N colors are selected as prevalent color by ranking, sum up their corresponding pixels, then calculate the coverage rate by dividing it with the total number of pixels contained in that image. This feature shows how often the most common color appears in the image. This is a simple test that may find solid backgrounds (Liu et al. 2010a; Jithesh et al. 2003; Liu et al. 2010b; Wang et al. 2010; Liu et al. 2010a).
- *Number of the colors*: Number of colors which are used for presenting an image can be computed directly from the pixel grid (Liu et al. 2010a,b; Wang et al. 2010).
- *Average Color*: It computes the average red, blue and green color values for the image. Average *RGB* represents the average values in R, G and B channel of each pixel in an image (Jithesh et al. 2003; Mehta et al. 2008; Zhen et al. 2009).
- *Color Variance*: Color variance can be computed directly from the pixel grid (Liu et al. 2010a,b; Wang et al. 2010).
- *Homogeneity*: Measure of the brightness variation within the image (Gargiulo and Sansone 2008; Gargiulo et al. 2008).
- *Spatial Correlogram*: Color correlogram of an image is a table indexed by color pairs, where the k_{th} entry for (i, j) specifies the probability of finding a pixel of color j at a distance k from a pixel of color i in the image. A color correlogram expresses how the spatial correlation of pairs of colors changes with distance. A color histogram captures only the color distribution in an image and does not include any spatial correlation information (Gao 2009; Gao et al. 2009, 2010).
- *Entropy*: An index of the brightness variation among the pixels in an image (Gargiulo and Sansone 2008; Gargiulo et al. 2008).

4.2.2 Edge

Edge detection is very important in digital image processing since edges are one of the most important features that play a critical role in presentation of an image. Edge is the boundary of the target and the background and only when obtaining the edge we can differentiate the

target and the background. That is why different aspects of edges are used to discriminate a group of images from another. Some edge-related features are listed here.

- *Edge Detection*: Edge detection is a fundamental tool in image processing and computer vision, particularly in the areas of feature detection and feature extraction, which aim at identifying points in a digital image at which the image brightness changes sharply or formally, it has discontinuities (Jithesh et al. 2003).
- *Gradient Orientation Histograms*: It is used in computer vision and image processing for the purpose of object detection. The technique counts occurrences of gradient orientation in localized portions of an image. The distribution of gradient orientation may reveal the characteristics of texts. The distributions of gradient orientation for natural images appear more uniform and noisy than those of spam images. Orientation histogram feature provides the histogram of orientation of edges in the image (Gao et al. 2008, 2009, 2010; Nhung and Phuong 2007; Wang et al. 2007; Gao 2009).
- *Primitive Length*: A primitive is a continuous set of maximum number of pixels in the same direction that have the same gray-level. Each primitive is defined by its gray-level, length and direction. Primitive length uses lengths of texture primitives in different directions as texture description (Mehta et al. 2008).
- *Average Length of the Edges*: First the length of all detected edges in the image are calculated, then average of them are used as an image feature for more processing (Gao 2009; Gao et al. 2009, 2010).
- *Edge Frequency*: By an edge detector an edges of the image can be obtained from an original image (Mehta et al. 2008).
- *Number of Edge Pixels*: It is aimed at detecting large background components overlapping with characters and hiding them, and was defined as the relative number of edge pixels which lie inside character-like components of the binary image (Biggio et al. 2008).
- *Total Number of Edges*: Total number of edges which detected with an edge detector method is calculated as a feature (Gao 2009; Gao et al. 2009, 2010).

4.2.3 Properties

This kind of features which listed here are the general image feature. They can be easily obtained by one pass scanning of an image.

- *Image Size, Width, and Height*: They are three common features which their computing cost is too small, and they can be extracted easily without coding and decoding (Uemura and Tabat 2008; He et al. 2009; Dredze et al. 2007; Wang et al. 2010; Zhen et al. 2009).
- *Aspect Ratio*: The aspect ratio of an image is the ratio of the width of the image to its height (width/ height) (He et al. 2009; Wang et al. 2010).
- *File Size*: Size of an image file in *KB* is used. It is expected that a spam image has file size of less than 20 KB. More than 90% of spam image's file size is smaller than a normal image (Uemura and Tabat 2008; Dredze et al. 2007; Wang et al. 2010).
- *File Format*: The file format of the image (JPEG, BMP, PNG, and etc) can also be used as a feature to filter image spams. The idea is based on the extension, the actual file format (as determined from metadata) and whether they match (Jithesh et al. 2003).
- *Image Metadata*: All of the information contained in the image metadata, including whether the image has comments, number of images (frames), bits per pixel, progressive flag, color table entries, index value, transparent color, logical height and width, components, bands, etc. (Jithesh et al. 2003).

- *Bit Depth*: Bit depth quantifies how many unique colors are available in an image's color palette in terms of the number of 0's and 1's, or bits, which are used to specify each color. This does not mean that the image necessarily uses all of these colors, but it can specify colors with that level of precision instead (He et al. 2009; Wang et al. 2010).
- *File Name*: When a legitimate user attaches an image to an e-mail and transmits a message, it is assumed that the user does not transmit the same image to the same addressee several times. However, spammers transmit the same image several times because they send spam e-mails several times in large quantities. Actually, there exists some image spam that attached the image of the same file name (Uemura and Tabat 2008).

4.2.4 Texture

Texture can be considered to be repeating patterns of local variation of pixel intensities. Some methods which their output as feature based on texture feature are listed here.

- *Texture Features*: Because computer-generated graphics usually contain homogeneous color patterns, they contain almost no texture in fine resolution. To extract graphics features, first wavelet transformation applied on the input images. Then, texture features extracted in three orientations (vertical, horizontal, and diagonal) at fine resolution. If any of these extracted texture features falls below a predefined threshold, the image is likely to be a computer-generated graphic (Wu et al. 2005).
- *Autocorrelation*: Autocorrelation measures the coarseness of an image by evaluating the linear spatial relationships between texture primitives. Large primitives give rise to coarse texture and small primitives give rise to fine texture. If the primitives are large, the autocorrelation function decreases slowly with increasing distance whereas it decreases rapidly if texture consists of small primitives. If the primitives are periodic, the autocorrelation function increases and decreases periodically with distance (Mehta et al. 2008; Zhen et al. 2009).
- *Co-Occurrence Matrices*: Whether considering the intensity or gray-scale values of the image or various dimensions of color, the co-occurrence matrix can measure the texture of the image. Because co-occurrence matrices are typically large and sparse, often various metrics of the matrix are taken to get a more useful set of features (Mehta et al. 2008; Gargiulo et al. 2008; Gargiulo and Sansone 2008).
- *Local Binary Pattern (LBP)*: It is a simple yet very efficient texture operator which labels the pixels of an image by thresholding the neighborhood of each pixel with the value of the center pixel and considers the result as a binary number. Due to its discriminative power and computational simplicity, *LBP* texture operator has become a popular approach in various applications. It can be seen as a unifying approach to the traditionally divergent statistical and structural models of texture analysis. Perhaps the most important property of the *LBP* operator in real-world applications is its robustness to monotonic gray-scale changes caused, for example, by illumination variations. Another important property is its computational simplicity, which makes it possible to analyze images in challenging real-time settings (Gao 2009; Gao et al. 2009, 2010).
- *Geometric Moment*: Image moments are certain weighted (moments) averages of the image intensity of the pixels, or functions of those moments, usually chosen to have some attractive property or interpretation. When normalized, they can be considered as a probability distribution. The central moment is useful in representing the orientation of the image content (Mehta et al. 2008).

- *Symmetry*: Symmetry as a one of the image feature determine is an image symmetrical or not. Usually with a Boolean value (Gargiulo and Sansone 2008; Gargiulo et al. 2008).
- *Correlation*: An index of the correlation degree among the pixels (Gargiulo and Sansone 2008; Gargiulo et al. 2008).

4.2.5 Layout

These features obtained by analyzing the image in the point of layout or object view of an image. Because spam images are generated as mass, usually they have similar layout with fixed objects in it. Exploiting this feature is the target of identifying the similarity of images on aspect of image layouts.

- *Moment invariants*: Moment invariants are important shape descriptors in computer vision. Moment invariant produces a set of feature vectors that are invariant under shifting, scaling and rotation. The technique is widely used to extract the global features for pattern recognition due to its discrimination power and robustness. Their extraction is also based on gray-level image (Qu and Zhang 2009).
- *Eccentricity*: An approximate measure of eccentricity or elongation of an object in image (Mehta et al. 2008).
- *Local Invariant Feature*: To avoid near duplicate detection, spammers often change the layout of the advertisement and produce a series of variations. However, in these variations always there are some regions which will remain the same. Local invariant features are able to find correspondences in spite of large changes in viewing conditions, occlusions, and image clutter (Zuo et al. 2009).

4.2.6 Other

Some feature are not generally used or predefined before while they are defined by authors and are supposed to help the image spam detection.

- *Blocks Averaging*: It is based on composition. The primary step involved in this module is resizing the image. Then the images are partitioned into blocks, and represent the region arrangement of the image by the distribution of the uniformity of pixel values within each block. After partitioning the image, mean value of the block that represents the right and left extreme of the image and the center block of the image are considered for further processing because spam images have fewer contents than natural images. These selected blocks are assumed to represent the super blocks of the image. The uniformities in the blocks adjacent to the super blocks are evaluated to determine the feature points in an image (Soranamageswari and Meena 2010).
- *Compressibility*: When converting the contents of a spam e-mail to an image, it is necessary for the image to have a comprehensible size. In addition, the file size of spam images tends to be small. Therefore, it is assumed that their compressibility might be higher than conventional images. This shows that spam images are more simple images than other images (Uemura and Tabat 2008; Zhen et al. 2009).
- *Extent of Text Region*: The extent of text in the image is defined as the fraction of the total area of text regions within the whole extent of image (Dredze et al. 2007; Wu et al. 2005; Biggio et al. 2008).
- *Perimetric Complexity*: It is defined as the squared length of the boundary between black and white pixels in the whole image (the perimeter), then divided by the black area.

Perimetric complexity is defined for black and white images, as the squared length of the boundary between black and white pixels in the whole image (Gargiulo and Sansone 2008; Biggio et al. 2008).

Some of the content-based techniques do not use features of image content directly. But by analyzing the image content and exclusive approaches they try to detect spam images. In Nielson et al. (2008) authors have proposed a technique by applying existing software and decades-old technology. First all images are converted to JPEG format then by using JP2A the ASCII images of JPEG images are obtained. Then five attributes of the ASCII image computed to detect image spam: average run length, symbol spread, predominant symbol, sharp edges, the percentage of edges and ratio.

In Kim et al. (2010) authors proposed a novel solution, named as BLASTed Image Linkage (BASIL), to the near-duplicate image detection (NDID) problem by bridging two seemingly unrelated fields - Multimedia and Biology. They use three group feature (color-based, texture-based, and semantic features) and the popular gene sequence alignment algorithm in Biology, BLAST, to determine the similarity between two images for detection. In Cheng et al. (2010) authors have proposed a framework named BFMLC (Binary Filtering with Multi-Label Classification) to take both spam image filtering and user preferences into account. BFMLC framework can not only discriminate spam images from ham images, but also classify spam image as several predefined topics. It comprises feature representation, binary filtering and multi-label classification. Authors believe that because some of the extracted features may be irrelevant or redundant, feature selection is needed to select a subset of relevant features for building robust learning models. After feature extraction and feature selection, an image will be represented as a feature vector which the number of selected features determines the vector dimension.

4.3 Text based

4.3.1 OCR

Generally OCR is defined as translation of images including handwritten, typewritten or printed text into text. The purpose of the OCR algorithm is to convert images or scanned documents in paper format into electronic document that is editable and understandable by machine. A good OCR system should be able to recognise the characters in different fonts, styles and sizes. The early designed systems were able just to process one or two sets of characters in fixed type and size. But today modern character recognition systems using sophisticated techniques, have mission to recognising of complex characters and symbols with different sizes.

OCR mainly use two methods for recognition, matrix matching and feature extraction. Matrix matching or pattern matching is used to compare and check what the OCR device sees including handwritten, typed, or printed text with a library of known characters. Matrix matching is used in older OCR tools or in a bounded environment including a few numbers of character and styles with little or no variation within each style. Feature extraction does not rely on precise matching, it analyzes features such as open areas, closed shapes, diagonal lines, line intersections, etc. Feature extraction methods are much more versatile than matrix matching. Spam filters exploit OCR techniques to extract text of image for further processing. After text extraction, OCR based methods using traditional spam filter techniques to analyze text for finding special keywords related to image spam. Then image can be identified as image spam or not. Sometimes it can be successful but recently almost all of the image

spammers use a variety of obfuscation techniques to obfuscate image spam for circumvents anti spam filters.

At the first view, OCR is one of the best choices for image spam filtering, but two big issues come along OCR which should be taken into account when the filtering is based on OCR. These two disadvantages cannot be ignorable (Fumera et al. 2007). First it imposes high computational cost to image spam filtering process, second OCR is vulnerable and spammer exploit a lot of tricks (As mentioned in “image spam tricks” part) to fool OCR. Although OCR overcomes some of them but already some others are really problematic and with their existence, OCR cannot work properly. Note that each time OCR is updated to overcome new obfuscation, probably this will lead OCR to a higher computational cost level. In 2005 and before no content obscuring techniques were used by spammers for text embedded into attached images (Wu et al. 2005). As investigated by authors in Biggio et al. (2007a) OCR tools can be effective for detecting image spam, for cases in which no content obscuring techniques are exploit by spammers.

We analyzed all important tricks used by spammer in “Image spam tricks” part. Moreover we proposed some new tricks which can be used by spammers in the future, for instance we believe that “patchy font” can be so problematic. Deliberate obfuscation makes it more difficult to determine which pixels are text and which are background color or noise. Spammers often obfuscate the text by using raised lettering, using different sizes and irregular fonts, blurring of text, rotating text, adding random colors, adding random shapes in backgrounds and so on (Soranamageswari and Meena 2010; Gao et al. 2008; Aradhye et al. 2005). Another reason for the failure of OCR is the lack of use of text in some kinds of image spams. As reported by [60] the lower TP rate of OCR based techniques originate from the lack of text in some spam images. Some examples are the images used for pornographic purpose, companies logos, shopping or even friend finding purposes. Proposed methods in newer researches do not use OCR (Soranamageswari and Meena 2010; Mehta et al. 2008; He et al. 2009; Gao et al. 2008; Aradhye et al. 2005; Biggio et al. 2007a,b).

Some papers send ball to opponent’s field, they do not try to recognize and catch the letters or words instead they try to realize existence of obfuscation. In other word as today obfuscation is used widely aimed at fooling OCR, they consider existence of obfuscation as proof for being spam. For example in Biggio et al. (2007b) and Biggio et al. (2007a) authors suggest a specific approach aimed at detecting the presence of noisy text due to the use of content obscuring techniques against OCR tools. In Aradhye et al. (2005), authors try to find text region instead of recognition of text and use this criteria for making decision whether the image is spam or not. Whereas OCR used by authors in Ma et al. (2006), it is obvious that their approach’s success rely on accuracy of OCR. In order to reducing the computational cost of OCR, the paper Fumera et al. (2006) proposes an approach to anti-spam filtering which exploits the text information embedded into images. Authors in Fumera et al. (2006) use hierarchical architecture for the spam filter aiming to reduce the high computational cost of OCR, note that it just reduce the number of OCR running not the cost of OCR running, so that it just impalement on the suspicious cases, not for all cases.

4.3.2 CAPTCHA

The paper Rusu and Govindaraju (2004) presents an application called CAPTCHA. CAPTCHA (Completely Automatic Public Turing test to tell Computers and Humans) was emerged to aim at distinguishing internet communications originating from humans from those originating from software robots, definition from Chew and Tygar (2004). For the first time Chew and Tygar created a new type of CAPTCHA, called image CAPTCHA Rusu and Govindaraju

(2004). We believe that the vulnerability of OCR made a situation for CAPTCHA emerging to be used as an image spam filtering technique. CAPTCHA approaches provide a barrier across the entrance of spam image emails. If the spam image email, pass through CAPTCHA it is identified as legitimate, and if it cannot pass CAPTCHA it can be considered as spam (Soranamageswari and Meena 2010; He et al. 2008). This filter approach change the content text to the image files and successfully separate human from machines. As mentioned before blurring of text outlines, construction of the image from multiple image layers, adding random noise, changing foreground colors, background colors, and rotating wavy text, animating, and etc which will not prevent the spam image from being read by human beings. Surprisingly this is the same phenomenon exploited by CAPTCHAs, but directly target spammers, rather than to deter their activity [33]. CAPTCHA belongs to the set of protocols called HIP (Human Interactive Proofs) which allow a person to authenticate as belonging to a select group, for example human as opposed to machine, adult as opposed to a child, etc. (Rusu and Govindaraju 2004).

Various CAPTCHAs can be designed and implemented like the ones proposed in Chew and Tygar (2004): naming images, distinguishing images, and anomalous images. In naming images some images provided to user and user should correctly type common term of the images, in this case user pass the test and identified as human. In distinguishing images two groups of images are provided to the user, these two groups can be the same or not, user should be able to say whether they are the same or not. In anomalous images some images are provided to user so that one is different from others, and user should be able to find the different one. The paper Fritsch et al. (2010) develops an approach to attack image recognition CAPTCHA called PixelMap and give several suggestions to improve future image recognition CAPTCHAs. In this method there are three steps, in first step they calculate PixelMap for the original image dataset, in the second step they calculate PixelMap for each image in the test and finally in third step they compare PixelMaps obtained in step one and two.

Table 1 shows which detection method can be used for detecting a specific image spam trick. Finally, Fig. 17 shows a complete classification of image spam filtering techniques.

Table 1 Classification of image spam tricks versus image spam detection techniques

	Header based	Content based	Text based
Obfuscation			✓
Text only		✓	✓
Multiple sections		✓	✓
Hand-written			✓
Wild background			✓
Geometric variance		✓	
Multi-frame animated Gifs		✓	
Cartoon		✓	✓
Recoloring		✓	
Obfuscate and shorten URLs			✓
Forge header information	✓		
Template-driven	✓	✓	
Emerge new source	✓		
People's computers	✓		
Text image			✓
You can unsubscribe!			✓
Natural image		✓	✓
Patchy font		✓	✓

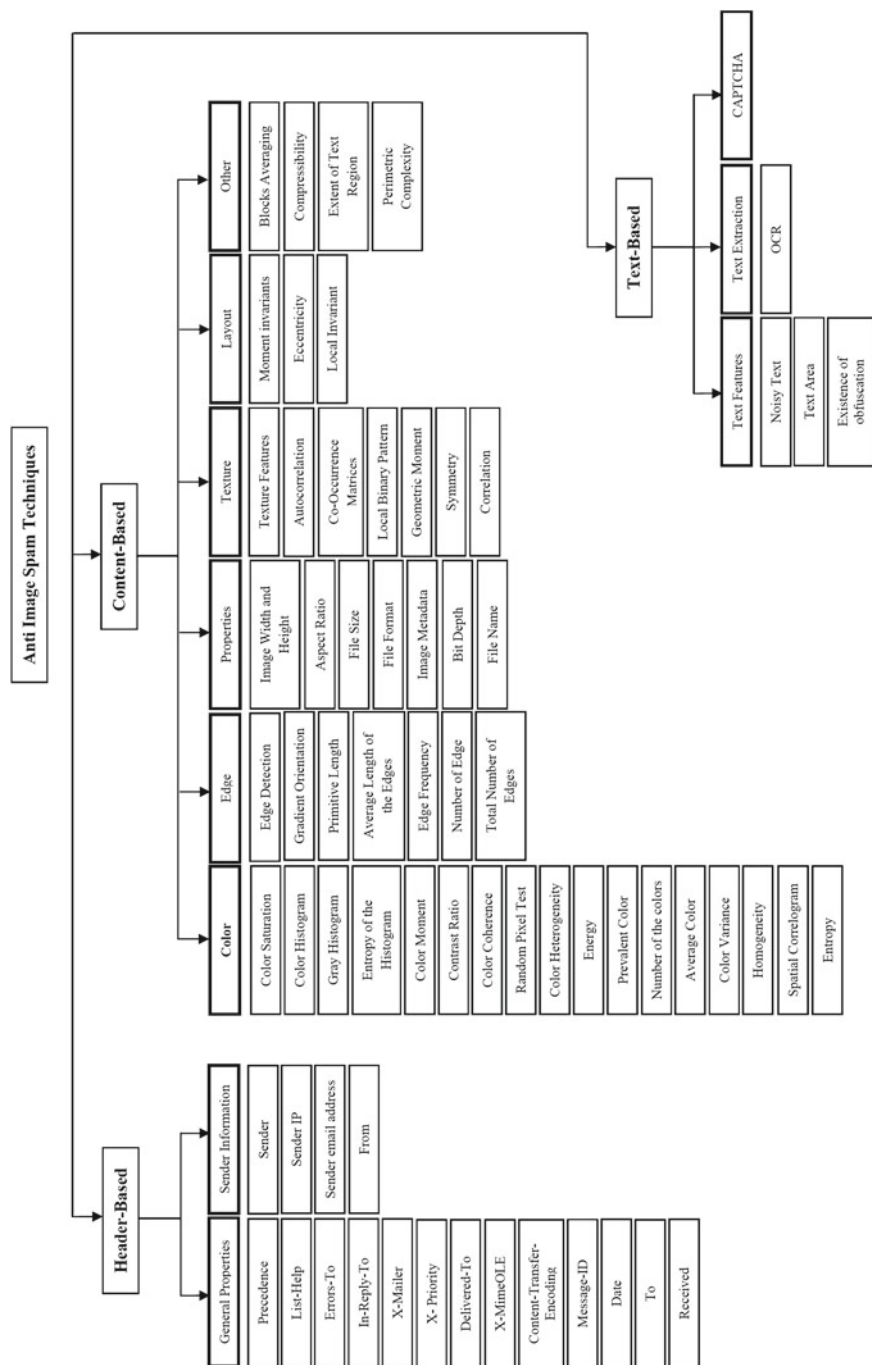


Fig. 17 A complete classification of the image spam filtering techniques

5 Data set and evaluation

The presented methods in any field definitely need a creditable data set to test upon it, and examine efficiency in compare to other's related work. Since email communications are inherently private, creating a proper data set for spam image emails is very difficult (Klangraphant and Bhattarakosol 2010). There is no universal corpus for image spam. We investigate other approaches in this literature, some works build their own data set for evaluation (Dredze et al. 2007; Wu et al. 2005; Fumera et al. 2006; Cheng et al. 2010). Some others use data sets which are available on the internet for this purpose (Mehta et al. 2008; He et al. 2009; Liu et al. 2010b). The work Soranamageswari and Meena (2010) has used 5000 images which are randomly collected from spam archive data set.

As we mentioned before building a data set (both spam and legitimate images) is difficult, it is even worse when we are speaking about legitimate images, because they are completely personal. Some researchers Wu et al. (2005); Zuo et al. (2009) collect their legitimate images by searching Google engine using keywords: Flower, baby, photo, Boy, and etc. Some others use the emails that authors or limited group of people received during a certain period (He et al. 2009; Gao 2009; Gao et al. 2008, 2009, 2010). For example in Gao et al. (2009, 2010) the authors collect data set images from spam emails received by 10 graduate students in their department between Jan 2006 and Feb 2009. Some others use the data set used by others, for example paper Cheng et al. (2010) use data set of paper Dredze et al. (2007) and paper Zhen et al. (2009) use data set that before used in paper Nhung and Phuong (2007). The spam images are created automatically using random techniques, so naturally they are similar to each other. Existing methods act in different ways, some illuminate them, and some preserve them. These different reactions are based on different aims and approaches of the methods, for example some like Dredze et al. (2007) remove similar images because they are trying to create a unique data set. The paper Cheng et al. (2010) remove images smaller than 10×10 pixels since those images are often used as blank spaces in HTML documents and also they remove spam images without any embedded text. They believe that spam images without any embedded text seems more reasonable to be as ham. Table 2 shows the details of data set used by researchers. But today in real word we encounter with spam images without text embedded in, like images for pornographic or shopping or even friend finding purposes. All the approaches mentioned above have used poor data sets, because data set cannot be selected randomly or without any wisdom. Also we cannot exclusively rely on the emails received by a group including some people, we list some reasons to called all above data sets unreliable here:

- There is huge amount of spam email sources that send thousands spam each day to address various purposes so limited number of people receives special kinds of spam.
- The concept of being spam is different for each people, for example scholarship emails are useful for some persons but bothersome for others.
- In building these data sets authors paid no attention to types of image spams contributed in data set.
- A proper data set should be able to reflect real world of spam emails, not represent just a specific group (for example belonging to some people or subject).

We believe that the contribution of image spam types is a critical factor that should be considered. A proper data set should be able to reflect the real world of spam images. M86 security lab provides statistic report on week ending December 19, 2010, see in Fig. 18. Figure 19 shows the Symantec Global internet Security threat report which is provided by

Table 2 Details of data set used by researchers

Image spam	Legitimate image		Used in
	Collected from	Amount	Amount
www.spamarchive.org	Personal data set	8,444	2021
Spam messages arriving at an email server	e-card collected from friends and free websites—random images from CorelDraw collections—Google-image search engine	411	1310
Mailboxes of some users of the studenti.unina.it mail server during three years (2005–2007)	Mailboxes of some users of the studenti.unina.it mail server during three years (2005–2007)	1802	151
www.spamarchive.org —Personal spam	Personal data set—Search in Internet	11,831	3,784
Data set of Dredze et al. (2007)	Data set of Dredze et al. (2007)	3,114	1,699
The mailboxes of all the students of the University of Naples Federico during three years (2005–2007)	Federico during three years (2005–2007)	3731	178
Personal data set—Internet source including http://unspammable.xtdnet.nl/2003/attachment.html	Personal data set—Internet source including http://unspammable.xtdnet.nl/2003/attachment.html	1,492	1,159
http://ce.dice.unica.it/spam-images.zip	http://ce.dice.unica.it/spam-images.zip		Nielson et al. (2008)
Data set of Nhung and Phuong (2007)— www.spamarchive.org	Data set of Nhung and Phuong (2007)— www.spamarchive.org	11,846	2,006
Received by 10 graduate students in authors department between Jan. 2006 and Mar. 2009	Downloading from photo sharing site Flickr.com image search engines such as Microsoft live image search (http://www.live.com/?scope=images)	1,190	Zhen et al. (2009)
www.spamarchive.org —Personal spam	Authors homegrown legitimate images—Data set of Dredze et al. (2007)	3,885	Gao (2009)
Not mentioned	Not mentioned	3,711	Zuo et al. (2009)
TREC data set available at http://trec.nist.gov/data/spam.html	TREC data set http://trec.nist.gov/data/spam.html	13,401	Krasser et al. (2007)
			Liu et al. (2010b)

Table 2 continued

Image spam	Legitimate image		Used in
	Collected from	Amount	
Data set of Gargiulo et al. (2008)—Data set of Dredze et al. (2007)—Real spam mails			
Spam images by the authors since January 2006—Data set of Dredze et al. (2007)	Data set of Dredze et al. (2007)—Randomly downloaded from Flickr.com	11,846	
http://prag.dice.unica.it/n3ws1t0/eng/spamRepository	Data set of Dredze et al. (2007)	2,006	Biggio et al. (2008)
Several hundred e-mails from spam box	http://prag.dice.unica.it/n3ws1t0/eng/spamRepository		
www.spamarchive.org —Personal spam	Google-images search engine	1,245	Aradhye et al. (2005)
Image spams received by the authors during 6 months	Personal data set—Google-images search engine	12,483	Wang et al. (2010)
www.spamarchive.org —Personal spam	Randomly selected images from Flickr.com	928	Gao et al. (2008)
www.spamarchive.org —Personal spam	Personal data set	12,742	Dredze et al. (2007)
Seven different email accounts during three months (Dec. 2006 to Feb. 2007). http://www.cs.princeton.edu/cass/spam/spam_bench/	Personal data set	11,831	Liu et al. (2010b)
Data set of Wang et al. (2007)—Two different email account	Downloaded from PBase and Photonet, and the COREL stock photo collection	1,071	Wang et al. (2007)
	Google-images search engines	1,977	He et al. (2009)

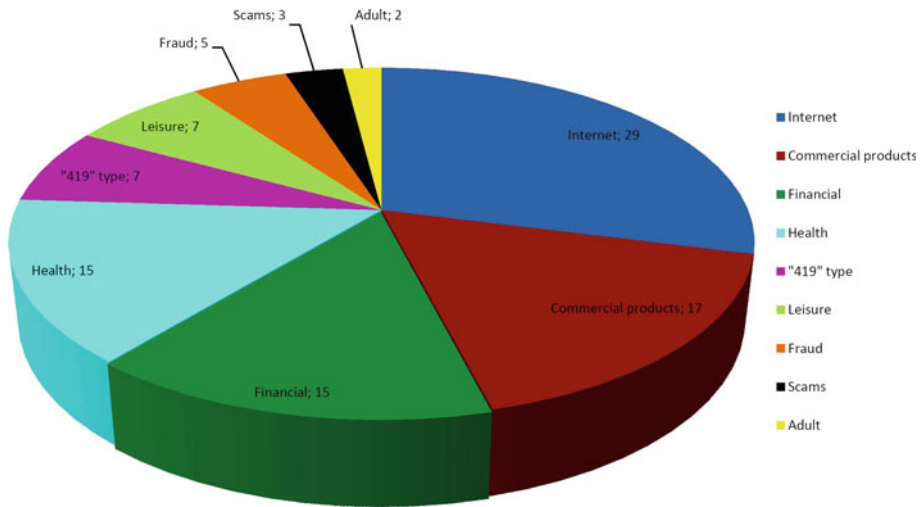


Fig. 18 Statistical classification of image spams provided by secure web gateway (internet security and email security solutions, available at <http://www.m86security.com>)

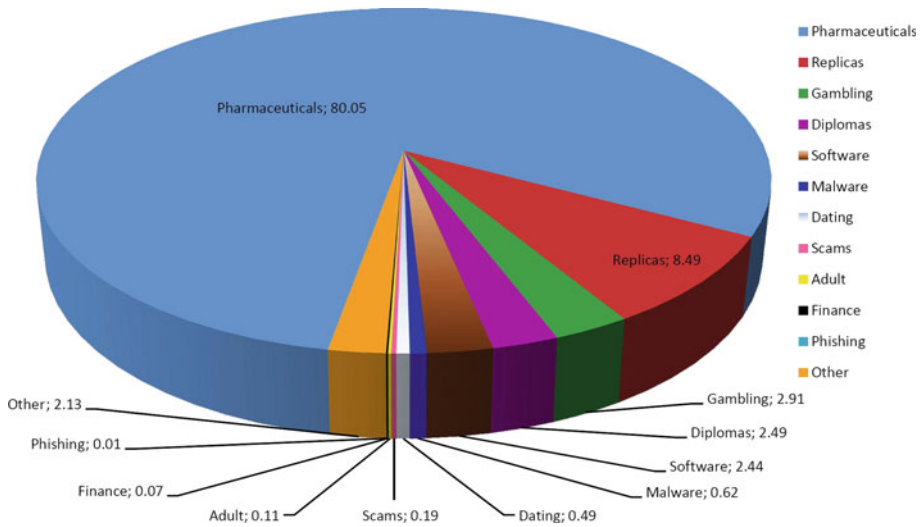


Fig. 19 "Symantec global internet security threat report" provided by Symantec Company in April 2010 (SYmantec—Antivirus, Anti-Spyware, available at <http://www.symantec.com>)

Symantec Company in April 2010 (SYmantec—Antivirus, Anti-Spyware, available at <http://www.symantec.com>).

As we mentioned before a reliable data set should reflect the real world of image spam, for example according to Symantec report the most common type of spam detected by Symantec in 2009 was related to internet-related goods and services with 29% of all spam. It means the spammer trend to internet-related goods and services more than other topics. Internet-related goods and services images should constitute 29% of data set images.

5.1 False positive/negative and true positive/negative

Because different images have similar (even same) visual features, filtering technologies of image spam may be wrong in detection. So it should be criteria for evaluation of various proposed method. False Positive (FP), False Negative (FN), True Positive (TP), and True Negative (TN) are four quantities which usually adopted by image spammer researcher to compare the accuracy of different approaches. Binary classifiers are normally used for the evaluation of the image spam filtering techniques in classification of images as spam image or non-spam image. Binary classification is the task of classifying the members of a given set into two groups on the basis of whether they have some property or not. To measure the performance of this binary classifier test for images as group of spam images and non-spam images if the image detected as spam it means the result of our image spam detection test is positive and if image detected as non-spam image the test result is negative. So, they are defined as below:

- FP: the false positive rate indicates the portion of the non-spam images wrongly being classified as spam images.
- FN: the false negative rate indicates the portion of the spam images wrongly being classified as non-spam images.
- TP: the true positive rate represents the portion of the spam images correctly being classified as spam images.
- TN: the true negative rate represents the portion of the non-spam images correctly being classified as non-spam images.

It should be said that most often a spam detection system would prefer to work with false positive rate. Because people prefer having spam images in their inbox inevitably than having their non-spam images in spam box. So a system that is able to produce very few false positives means that it has a very good precision in detection. Table 3 shows the details of the most common formulas which are used by researchers for evaluation of their proposed methods.

6 Open issues

Anti spam researchers expect spammers embed unsolicited materials into images or text format in body of emails. Although spammer obtained remarkable success in use of image spam but anti spam systems did not lose the game completely, and achieve some periodic success. This is enough to spammers realize filters had some problems with image spam. As the never-ending Cat and Mouse game, the story repeat and spammer came out with a new approach called attachment based spam as reported in white paper (<http://www.gfi.com/>). In this new trend of spam spammer exploit some credible and high-usage file format which specially use in official, business, and academic fields like pdf, doc, docx, ppt, and etc. Motivation of this new trend:

- Because each important document over internet is in one of these formats, people fear to delete and reject such emails without reading or opening.
- Anti spam systems extremely worry about false positive rate, even minimal amount false positive is major problem because users lose important documents. It can be considered as a bold weakness for anti spam systems.
- It cause filtering and detection so hard and costly, can destroy circumstances as well as image spam even more.

Table 3 Evaluation criteria for image spams

Measure	Formula	Description	Used in
Accuracy (ACC)	$\frac{(TP+TN)}{(TP+TN+FP+FN)}$	Ratio between the number of correctly identified image spam and legitimate image to the total number of images	Qu and Zhang (2009), Gargiulo et al. (2009), Liu et al. (2010a), Cheng et al. (2010), Gargiulo et al. (2008), Zhen et al. (2009), Gao (2009), Zuo et al. (2009), Youn and McLeod (2009), Liu et al. (2010b), Gao et al. (2008), Dredze et al. (2007), Soranamageswari and Meena (2010)
False positive rate (FPR)	$\frac{(FP)}{(FP+TN)}$	Ratio of spam images detected incorrectly	Qu and Zhang (2009), Liu et al. (2010a), Nielson et al. (2008), Gao et al. (2009), Liu et al. (2010b), Krasser et al. (2007), Fumera et al. (2006)
Positive rate	$\frac{(TP)}{(TP+FN)}$	Ratio of spam images detected correctly	Gao et al. (2008), Wu et al. (2005), Wang et al. (2010)
(TPR or spam recall)	$\frac{(TP)}{(TP+FP)}$	Ratio of spam images detected correctly	Nhung and Phuong (2007), Gargiulo and Sansone (2008), Nielson et al. (2008), Kim et al. (2010), Gao et al. (2009)
Spam precision	$\frac{(TP)}{(TP+FP)}$	That actually are image spam That actually are image spam	Krasser et al. (2007), Kim et al. (2007), Gao et al. (2008), Soranamageswari and Meena (2010)
F1-measure	$2 \times \frac{(Precision \times Recall)}{(Precision + Recall)}$	Weighted average of the precision and recall	Qu and Zhang (2009), Cheng et al. (2010), Gargiulo and Sansone (2008), Kim et al. (2010), Kim et al. (2007), [33], Soranamageswari and Meena (2010)
ROC curve	True positive rate plotted against false positive rate	Relative Operating Characteristic curve, a comparison of TPR and FPR as the criterion changes	Chen and Zhang (2009), Gargiulo and Sansone (2008), Zhen et al. (2009), Nielson et al. (2008), Krasser et al. (2007), Fumera et al. (2006), Biggio et al. (2008)

- Facility of sending large file cause spammers embed all desirable materials into attachment file, and this could result in reducing the quality and bandwidth of server.

Spam will continue to become a nightmare for institutes and end-users. Spammers are trying hard to be one step ahead and it is so bad because in this condition even if the techniques and tricks which exploit by spammer can be prevented and solved by anti spam researchers, anti spam researchers lose the time. Additional time provide spammers adequate opportunity to come to the field with something new even in help of new experience. We believe that attack is the best form of defense and must do not wait for spammer to restart the game in new round.

Spam filtering techniques should not be stand alone and ad-hoc in the battle against image spam. Two important aspects that is necessary to get into considered for achieving remarkable success. Human Analysis: Spam researcher can not ignore the critical role of human and rely solely on OCR technology to combat image spam, because solely machine solutions are rarely foolproof. Human analysis is required to improve machine performance and correct inevitable errors. Progressive Technology: Image spam is last round in the fight between spammers and email users. As it is a never-ending cat and mouse game and spammers constantly change their tactics and techniques, Anti spam methods can rely on dynamic and flexible designs to provide the capabilities and meet the requirements of tomorrow's fight. Stability, flexibility, and scalability are some critical features for an efficient anti spam. We should not forget old kinds of spam emails such as text based emails included hyperlinks as we are treat and fight against new techniques like image spam.

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